

Day 21 of training

Conditional Rendering in React

Definition:

Conditional Rendering in React is a technique where components or elements are displayed based on a condition. It's similar to using if-else statements in JavaScript but for UI rendering.

- It allows React apps to show different content depending on state, props, or other conditions.
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Key Points:

1. Used to render elements dynamically based on conditions.
2. Helps make the UI interactive and responsive.
3. Can be done using if-else, ternary operator, logical &&, or switch-case.
4. Works in functional and class components.

Methods of Conditional Rendering

1. Using if-else inside component
2. Using Ternary Operator
3. Using Logical && Operator-(If `message` is empty or false, nothing is rendered.)
4. Using Switch-Case (for multiple conditions)

Advantages of Conditional Rendering:

1. Makes UI **dynamic and interactive**.
2. Avoids writing **duplicate components**.
3. Works with **state, props, or any condition**.
4. Easy to maintain and understand.

Real-Life Examples / Use Cases:

- **Login System:** Show “Login” button if the user is logged out, and “Logout” button if logged in.
- **Notification Messages:** Show messages only if there is new data.
- **E-commerce Site:** Display “Add to Cart” or “Out of Stock” based on product availability.
- **Form Validation:** Show error messages only when input is invalid.

Note – In my project chef cloude I used conditional rendering on add ingredient button and form when I type something then it show ingredient list and recipe